AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 22 · STUDENT LESSON

Pooling and spatial information

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

For a 3 by 3 input and a 2 by 2 window with stride 2 and no padding, how many valid window positions exist?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Compute max and mean pooling under an explicit window rule. Try the recall before opening notes.

Your learning target

I can compute max and **mean pooling** under an explicit window rule.

I can use collisions to explain lost **spatial information**.

I can test boundary and movement claims with revealing examples.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Regularisation and early stopping. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Compute max and **mean pooling** under an explicit window rule

Use collisions to explain lost **spatial information**

Test boundary and movement claims with revealing examples

Retrieve what you already know

> For a 3 by 3 **input** and a 2 by 2 window with stride 2 and no padding, how many valid window positions exist?

My starting explanation

This week's end goal

By the end, I can explain pooling and **spatial information** using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Pooling and spatial information case. Separate observations, given facts and assumptions before choosing an explanation.

A fictional single-channel feature map has rows [1, 4, 2, 0], [3, 2, 5, 1], [0, 6, 1, 3] and [2, 4, 2, 7]. Use a 2 by 2 window, stride 2, no padding and only complete windows. Coordinates are zero-based.

The four starts are [0, 0], [0, 2], [2, 0] and [2, 2].

Max pooling keeps the largest value in each window. **Mean pooling** adds the four values and divides by 4. Neither operation learns weights in this lesson.

These are fixed summaries of the incoming feature map, unlike the trainable kernel parameters introduced last week.

Window start	Values	Maximum	Mean
[0, 0]	1, 4, 3, 2	4	2.5
[0, 2]	2, 0, 5, 1	5	2
[2, 0]	0, 6, 2, 4	6	3
[2, 2]	1, 3, 2, 7	7	3.25

The max **output** is [[4, 5], [6, 7]], and the mean **output** is [[2.5, 2], [3, 3.25]]. Both have shape 2 by 2. Sixteen values become four, but the four **output**s still preserve the coarse arrangement of the four regions.

List the given facts and one fact that is not supplied.

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Pooling and spatial information. Connect each labelled input, deciding step and result.

This code assumes a nonempty rectangular numeric map at least 2 by 2. It uses only complete 2 by 2 windows, stride 2 and no padding.

It returns two separate summaries of the same map; it does not apply one pooling operation after the other.

```
def pool2(image):
    h, w = len(image), len(image[0])
    if h < 2 or w < 2:
        raise ValueError("Need a complete 2 by 2 window")
    maxima, means = [], []
    for r in range(0, h - 1, 2):
        max_row, mean_row = [], []
        for c in range(0, w - 1, 2):
            values = [image[r+i][c+j]
                      for i in range(2) for j in range(2)]
            max_row.append(max(values))
            mean_row.append(sum(values) / 4)
        maxima.append(max_row)
        means.append(mean_row)
    return maxima, means

x = [[1, 4, 2, 0], [3, 2, 5, 1],
     [0, 6, 1, 3], [2, 4, 2, 7]]
```

```
assert pool2(x) == ([[4, 5], [6, 7]],
                    [[2.5, 2], [3, 3.25]])
```

For this complete-window rule the **output** shape is floor((H - 2) / 2) + 1 by floor((W - 2) / 2) + 1. A 3 by 3 map produces just one **output**; its last row and last column are not read.

Do not silently invent a partial-window rule or pad missing positions.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Pooling and spatial information. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Give two different 2 by 2 windows containing exactly one 4 and three zeros that have the same maximum and mean. Explain why either scalar alone cannot identify the position of the 4.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

> **Change one thing**

Change: Swap1 and 4 within the left window.

Predicted result

Prediction: Both pooled maps remain unchanged.

Why the result follows

Reason: Max and mean ignore the positions of values inside the same window.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Pooling and spatial information to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Max pooling	Replacing each window by its largest value
Mean pooling	Replacing each window by the arithmetic mean of its values
Spatial information	Information about where values occur and how positions relate
Pooling collision	Different inputs producing the same pooled values

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Pooling and spatial information, predict a step, then compare with its explanation.

A maximum does not identify its location

INPUT WINDOW ROWS	MAXIMUM	MEAN
[4 0] and [0 0]	4	1
[0 4] and [0 0]	4	1
[4 4] and [4 4]	4	4

The first two windows differ in the position of their nonzero value yet have the same maximum and mean. The third has the same maximum but a different mean. Each summary retains some distinctions and discards others.

Compare windows [[0, 0], [6, 0]] and [[0, 0], [0, 6]]. Compute both summaries. Can a deterministic head receiving only either scalar distinguish bottom-left from bottom-right?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Pooling and spatial information

Pool a small map and show the detail it cannot retain.

The facts we will use

Supplied facts

Feature map rows [1, 4, 2, 0],[3, 2, 5, 1]. Pool non-overlapping2×2 windows, stride2, no padding.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/22

Two adjacent pooling windows

	left0	left1	right0	right1
row0	1	4	2	0
row1	3	2	5	1

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/22

Pooling and spatial information

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Left window

Values1, 4, 3, 2: max 4, mean 2.5.

2. Right window

Values2, 0, 5, 1: max 5, mean 2.

3. Compare summaries

Max map [[4, 5]], mean map [[2.5, 2]], both shape1 x 2.

CHANGE ONE THING

Swap1 and 4 within the left window.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Left window	Values1, 4, 3, 2: max 4, mean 2.5.
2. Right window	Values2, 0, 5, 1: max 5, mean 2.
3. Compare summaries	Max map [[4, 5]], mean map [[2.5, 2]], both shape1×2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Pooling compresses each local window into a summary. Maximum keeps its largest value; mean keeps its average. Neither summary records where each value appeared.

Swapping values inside one window leaves both summaries unchanged here, so the pooled **output** cannot reconstruct the original arrangement. The chosen window, stride and boundary policy determine the **output** shape.

What this example does not establish

Keep the stated limit

Pooling is a declared aggregation, not automatic improvement; losing location can harm tasks that need precise positions.

Keep the stated limit

Read the labelled picture: Can the max map reveal where the 4 originally appeared? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Swap1 and 4 within the left window. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Pooling and spatial information claim and explain a repair using the supplied evidence.

Check the claim against the facts

Within one isolated window, rearranging its values preserves both its maximum and mean. But moving a value across a window boundary can change the **output** map. In a 2 by 4 map of zeros, put a single 8 at row 0 column 1.

Max pooling gives [[8, 0]]. Move that 8 one column right to column 2, replacing the old position with zero: the **output** becomes [[0, 8]]. Pooling therefore does not guarantee that a classifier ignores every small movement.

Negative values are valid **input**s. For [[-4, -2], [-7, -3]], the maximum is -2 and the mean is -4. Starting a maximum calculation at zero would incorrectly return 0.

Adding a zero to the window would also change its maximum: padding choices matter. Our examples use no padding.

Check the claim against the facts

Returning the position of a maximum can preserve extra location information, but it still does not restore the discarded values.

For example, [[5, 1], [2, 3]] and [[5, 0], [0, 0]] have the same unique maximum 5 at the top-left position. Even the maximum plus its position cannot tell which original window was used.

Locate the earliest unsupported step

Apply the lesson complete-window rule to [[1, 2, 99], [3, 4, 99], [99, 99, 99]]. Give the **output** shape and both **output**s. Explain why neither **output** includes 99.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Pooling and spatial information with support, then explain the deciding step in your own words.

For the lesson window starting at [2, 2], list the four values, sum, maximum and mean. State the complete pooled **output** shape.

Compare windows [[0, 0], [6, 0]] and [[0, 0], [0, 6]]. Compute both summaries. Can a deterministic head receiving only either scalar distinguish bottom-left from bottom-right?

Explain your prediction to a partner before comparing results.

Practise with less help: Prove that location can be lost

Use workbook W2 for the connected example “Pooling and **spatial information**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Pooling and **spatial information**. Proposed explanation: “**Max pooling** turns negative numbers into zero”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Can the max map reveal where the 4 originally appeared? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Pooling and spatial information. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Give two different 2 by 2 windows containing exactly one 4 and three zeros that have the same maximum and mean. Explain why either scalar alone cannot identify the position of the 4.

```
def pool2(image):
    h, w = len(image), len(image[0])
    if h < 2 or w < 2:
        raise ValueError("Need a complete 2 by 2 window")
    maxima, means = [], []
    for r in range(0, h - 1, 2):
        max_row, mean_row = [], []
        for c in range(0, w - 1, 2):
            values = [image[r+i][c+j]
                      for i in range(2) for j in range(2)]
            max_row.append(max(values))
```

```
            mean_row.append(sum(values) / 4)
        maxima.append(max_row)
        means.append(mean_row)
    return maxima, means

x = [[1, 4, 2, 0], [3, 2, 5, 1],
     [0, 6, 1, 3], [2, 4, 2, 7]]
assert pool2(x) == ([[4, 5], [6, 7]],
                    [[2.5, 2], [3, 3.25]])
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Pooling and spatial information. Explain why each step is justified.

Explain → apply → challenge

For the lesson window starting at [2, 2], list the four values, sum, maximum and mean. State the complete pooled **output** shape.

Apply the lesson complete-window rule to [[1, 2, 99], [3, 4, 99], [99, 99, 99]]. Give the **output** shape and both **output**s. Explain why neither **output** includes 99.

> A 2 by 4 map is zero except for 8 at row 0 column 1. Move the 8 to column 2 and set its old position to zero. Give the max and mean maps before and after. What claim does this refute?

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled pooling and **spatial information evidence** sheet to Image-pattern Network Investigation. Show how you can compute max and **mean pooling** under an explicit window rule.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Pooling can reduce the amount of data **process**ed by later layers. Whether the lost information is acceptable depends on the task.

A task about whether any large response exists in a region may benefit from a maximum; a task requiring exact positions may need the unpooled map or another representation. Do not assume smaller always means better.

A teammate says a 2 by 2 max-pooled map lets them reconstruct the original 4 by 4 map exactly and guarantees unchanged predictions after any one-pixel shift. Use one collision and one boundary example to challenge those claims.

Compare pooling choices with matched data splits and a declared evaluation rule. Fit model parameters on **training** data and choose alternatives using validation data, leaving final tests reserved.

Next week combines these components in a small image-model experiment.

A 2 by 4 map is zero except for 8 at row 0 column 1. Move the 8 to column 2 and set its old position to zero. Give the max and mean maps before and after. What claim does this refute?

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Pooling and spatial information case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

A maximum does not identify its location

INPUT WINDOW ROWS	MAXIMUM	MEAN
[4 0] and [0 0]	4	1
[0 4] and [0 0]	4	1
[4 4] and [4 4]	4	4

The first two windows differ in the position of their nonzero value yet have the same maximum and mean. The third has the same maximum but a different mean. Each summary retains some distinctions and discards others.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Pool a new map independently

Use rows [-4, -1, 2, 6], [-3, -2, 4, 0], [5, 1, -8, -2], [3, 7, -4, -6]. Under the lesson rule, give all four sums, the max map and the mean map. Explain why replacing a negative maximum by zero is an error here.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Use rows [-4, -1, 2, 6], [-3, -2, 4, 0], [5, 1, -8, -2], [3, 7, -4, -6]. Under the lesson rule, give all four sums, the max map and the mean map. Explain why replacing a negative maximum by zero is an error here.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Pooling and spatial information, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Pool a small map and show the detail it cannot retain.

By the end, I can	How I will show it
Compute max and mean pooling under an explicit window rule	Explain the deciding step, show a trace or test, and state one limit.
Use collisions to explain lost spatial information	Explain the deciding step, show a trace or test, and state one limit.
Test boundary and movement claims with revealing examples	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Pooling and spatial information without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Compute max and **mean pooling** under an explicit window rule

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Pool a small map and show the detail it cannot retain.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Max map [[4, 5]], mean map [[2.5, 2]], both shape1×2.

1. Left window

2. Compare summaries

3. Right window

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare summaries Max map [[4, 5]], mean map [[2.5, 2]], both shape1×2. Pooling compresses each local window into a summary.

Maximum keeps its largest value; mean keeps its average. Neither summary records where each value appeared.

Swapping values inside one window leaves both summaries unchanged here, so the pooled **output** cannot reconstruct the original arrangement. The chosen window, stride and boundary policy determine the **output** shape.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Swap1 and 4 within the left window. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Both pooled maps remain unchanged.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Both pooled maps remain unchanged. Max and mean ignore the positions of values inside the same window.

Repair a missing building block

Review Convolution as a local operation (ai-5/21). Goal: Compute a local weighted sum at each valid image position. Short practice: Why is flipping the kernel an implementation change here?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learned representations (ai-5/19). Goal: Trace an **input** through a hidden representation and a decision rule. Short practice: Why can no deterministic head using only the sum classify both correctly?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Regularisation and early stopping (ai-5/16). Goal: Separate a weight penalty from **prediction** error. Short practice: Why should the penalty value not be reported as validation error?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

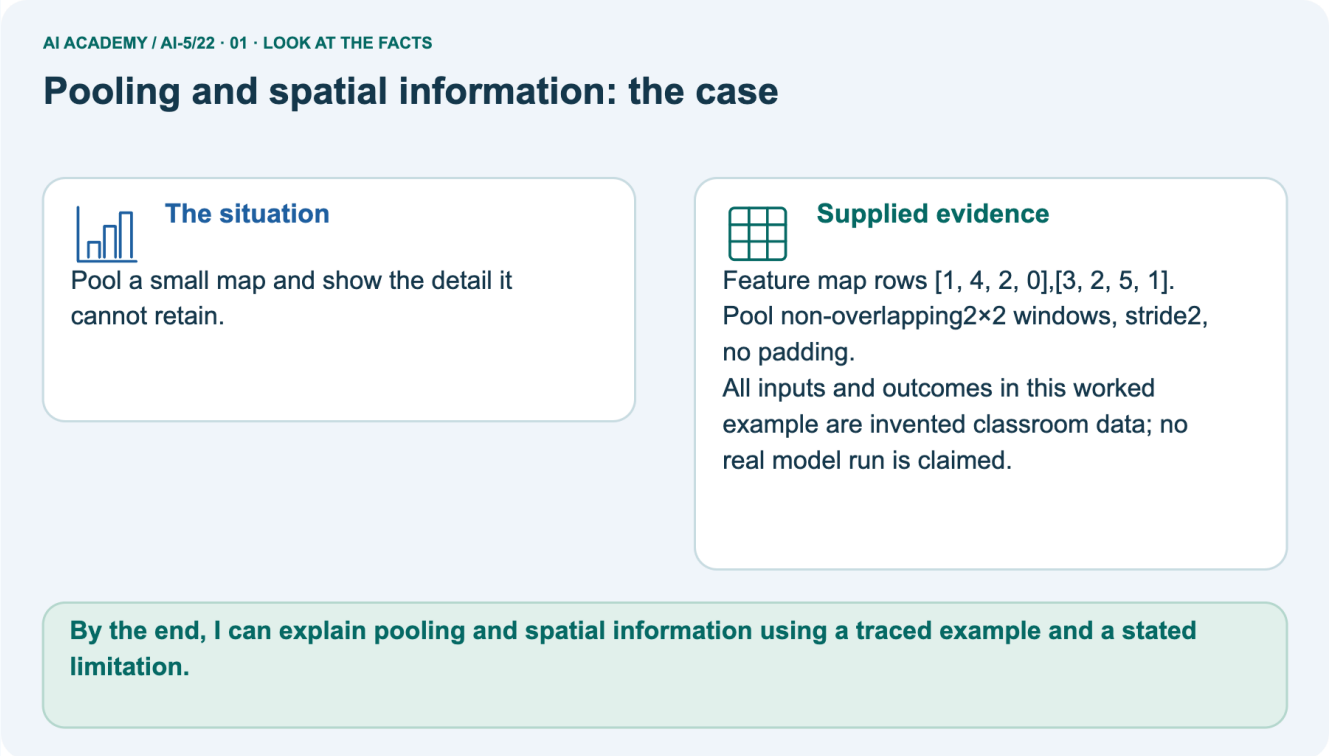
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Pooling and spatial information graphic using its labels, deciding evidence and relevant limitation.

01. Pooling and spatial information: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Pool a small map and show the detail it cannot retain. Supplied facts: Feature map rows [1, 4, 2, 0], [3, 2, 5, 1]. Pool non-overlapping2×2 windows, stride2, no padding.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Left window: Values1, 4, 3, 2: max 4, mean 2.5. → Right window: Values2, 0, 5, 1: max 5, mean 2. → Compare summaries: Max map [[4, 5]], mean map [[2.5, 2]], both shape1×2.

Point to each numbered stage and explain which recorded **input** it uses.

03. Two adjacent pooling windows

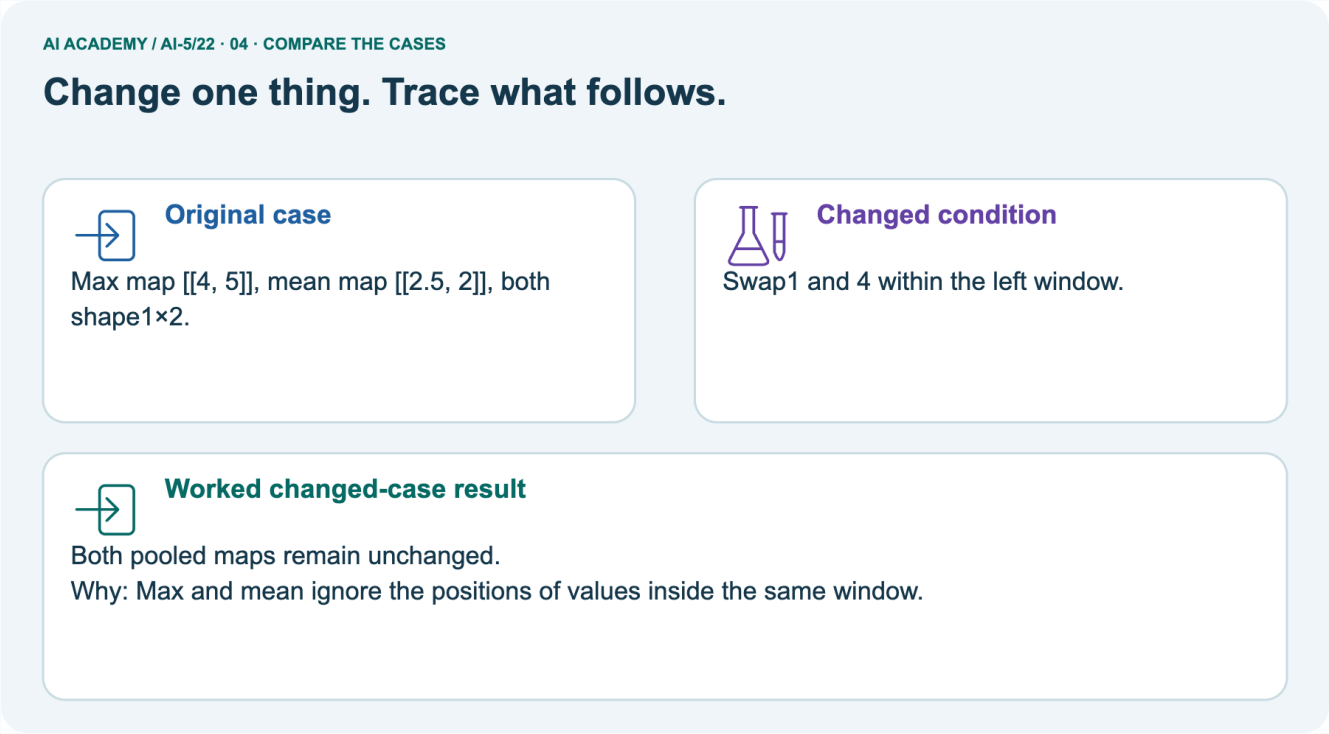


Follow the labelled picture with the matching explanation. Open the vector diagram

Two adjacent pooling windows. row0: left0 = 1, left1 = 4, right0 = 2, right1 = 0. row1: left0 = 3, left1 = 2, right0 = 5, right1 = 1.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



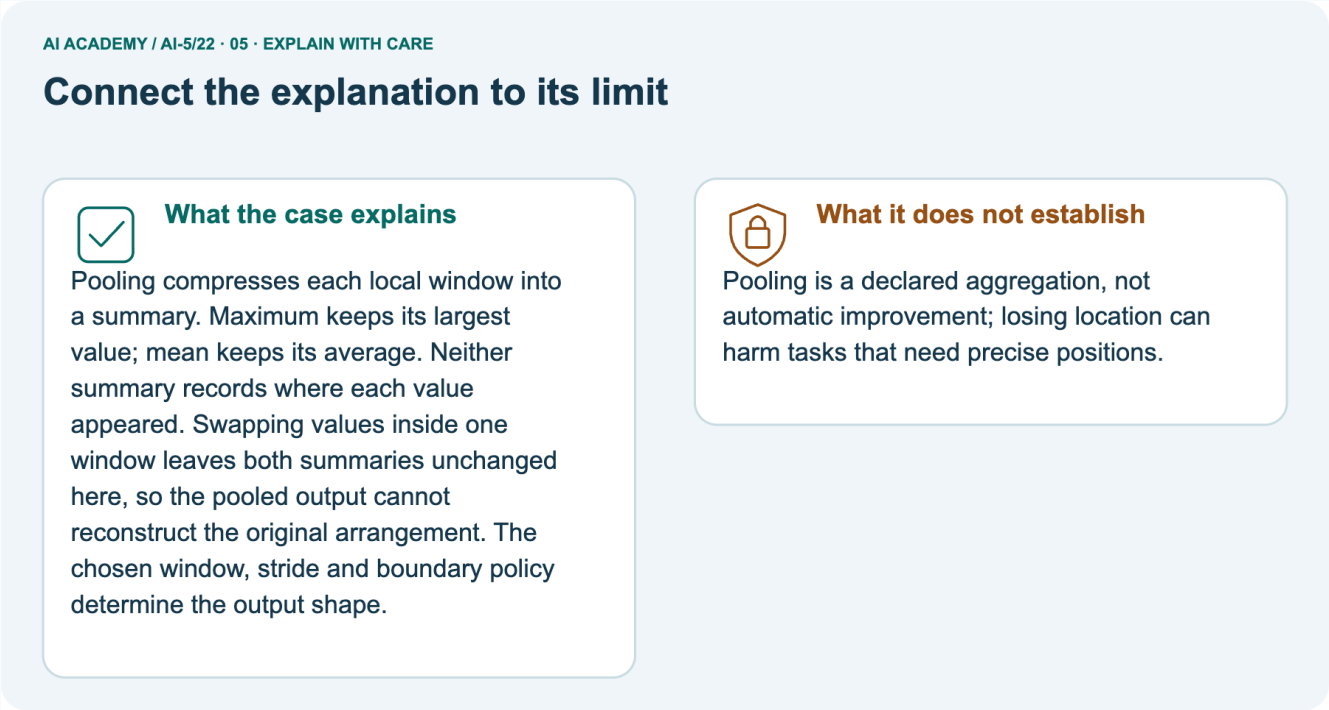
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Max map [[4, 5]], mean map [[2.5, 2]], both shape1×2. Changed condition: Swap1 and 4 within the left window. Expected result: Both pooled maps remain unchanged.

Reason: Max and mean ignore the positions of values inside the same window.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

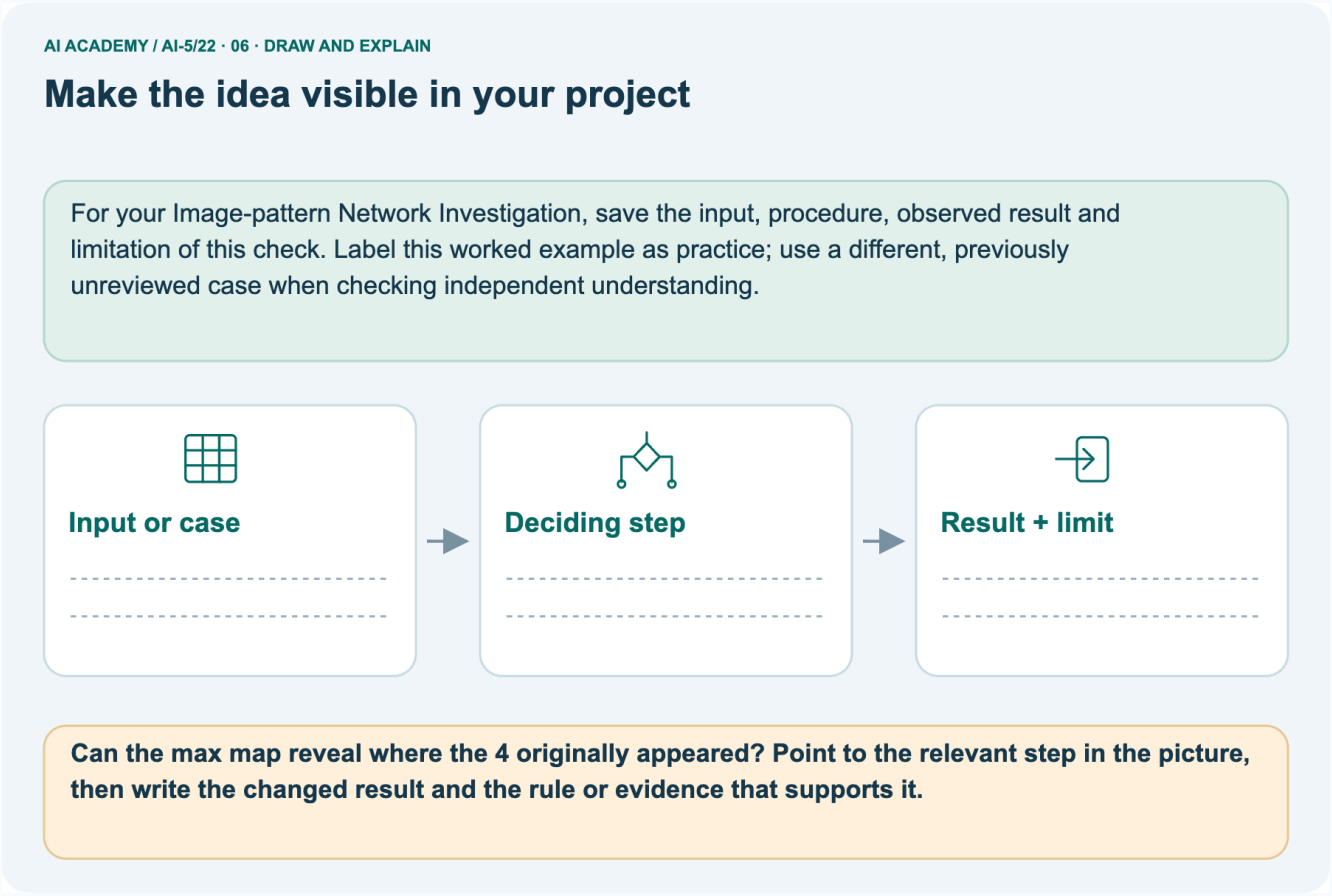
Pooling compresses each local window into a summary. Maximum keeps its largest value; mean keeps its average. Neither summary records where each value appeared.

Swapping values inside one window leaves both summaries unchanged here, so the pooled **output** cannot reconstruct the original arrangement. The chosen window, stride and boundary policy determine the **output** shape.

Limit: Pooling is a declared aggregation, not automatic improvement; losing location can harm tasks that need precise positions.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Can the max map reveal where the 4 originally appeared? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.